

PUF-Based UAV Authentication Protocol: Complete Four-Phase Architecture

Protocol Architecture

This protocol provides secure authentication for UAV swarm networks using Physical Unclonable Functions (PUFs). The complete authentication framework consists of four phases:

1. **Phase 1: Enrollment/Registration** - Initial setup with Ground Station in secure environment
2. **Phase 2: UAV-Ground Station Authentication** - UAV authenticates with GS when entering network
3. **Phase 3: UAV-to-UAV Peer-to-Peer Authentication** - Direct authentication between UAV pairs for collaborative missions
4. **Phase 4: Session Key Establishment** - Secure communication channel setup

Protocol Entities

- **Ground Station (GS):** Trusted authority managing UAV enrollment and credential distribution
- **UAV_i, UAV_j:** Autonomous aerial vehicles equipped with PUF hardware
- **PUF:** Physical Unclonable Function generating unique challenge-response pairs

Notation

Symbol	Description
UAV_i	i-th Unmanned Aerial Vehicle
ID_i	Permanent identity of UAV_i
TID_i	Temporary identity (for anonymity)
C_i	PUF challenge for UAV_i
$R_i = \text{PUF}_i(C_i)$	PUF response
$h(\cdot)$	Lightweight hash function (e.g., SPONGENT-160)
$a \oplus b$	XOR operation
$a b$	Concatenation
N_i	Random nonce generated by UAV_i
T_i	Timestamp
SK_{ij}	Session key between UAV_i and UAV_j
h_i	Helper data for error correction (BCH)
$\text{MAC}_K(M)$	Message authentication code of M with key K
$E_K(M)$	Encryption of message M with key K

1][2][3

Phase 1: Enrollment/Registration

Environment: Secure facility with tamper-proof communication channels

Purpose: Initialize UAVs and establish baseline credentials in Ground Station database

Step 1.1: PUF Interrogation

For each UAV_i to be enrolled:

GS Operations:

1. Generate a set of random challenges $\{C_i^{(1)}, C_i^{(2)}, \dots, C_i^{(n)}\}$ where $n \geq 12$
2. Send each challenge $C_i^{(k)}$ to UAV_i via secure channel

UAV_i Operations:

1. For each received challenge $C_i^{(k)}$:
$$R_i^{(k)} = \text{PUF}_i(C_i^{(k)})$$
2. Return response $R_i^{(k)}$ to GS

Security Note: This phase must occur in controlled environment. Physical security prevents CRP leakage.[5][1]

Step 1.2: Error Correction Setup

For each challenge-response pair, GS generates helper data for BCH error correction:

$$h_i^{(k)} = \text{BCH_encode}(R_i^{(k)}) \oplus R_i^{(k)}$$

Where BCH(255, 131, 18) can correct up to 18 bit errors ($\approx 7\%$ BER).[6][7]

Helper Data Properties:

- Public information (can be stored insecurely)
- Does not reveal $R_i^{(k)}$
- Enables reliable PUF response reconstruction under noise[8]

Step 1.3: Identity Assignment

GS generates for each UAV_i:

1. **Permanent Identity:** ID_i - unique 64-bit identifier

2. **Temporary Identity:** $TID_i = h(ID_i || \text{random_seed})$ - 64-bit pseudonym
3. **Registration Token:** $\tau_i = h(R_i^{(1)} || ID_i || T_{\text{reg}})$ where T_{reg} is registration timestamp

Step 1.4: Database Storage

GS Database Entry for UAV_i:

```
Entry_i = {
  ID_i,           // Permanent identity
  TID_i,          // Current temporary identity
  {(C_i^(1), h_i^(1)), ..., (C_i^(n), h_i^(n))}, // CRPs with helper data
  Status: "Registered", // Operational status
  T_reg,          // Registration timestamp
  Peer_credentials: {} // Empty, populated in Phase 2
}
```

UAV_i stores in non-volatile memory:

```
{TID_i,  $\tau_i$ , GS_public_params}
```

No raw PUF responses stored anywhere - they are regenerated on-demand.[4][1]

[9][10][1][5]

Phase 2: UAV-Ground Station Authentication

Environment: UAV in operational field, communicating with GS via wireless channel

Purpose: Authenticate UAV with GS upon network entry, obtain credentials for peer-to-peer authentication

Threat Model: Adversary can eavesdrop, modify, replay messages. Cannot clone PUF or break hash functions.[11][1]

Message 1: Authentication Request

UAV_i Operations:

1. Generate fresh timestamp T_1 and nonce N_1
2. Compute authentication hash:
$$\alpha_1 = h(\text{TID}_i \| T_1 \| N_1 \| \tau_i)$$
3. Transmit to GS:
$$M_1 = (\text{TID}_i, T_1, N_1, \alpha_1)$$

Message Format: 64 + 32 + 128 + 160 = 384 bits

Message 2: Challenge Issuance

GS Operations upon receiving M_1 :

1. **Verify Freshness:**
$$|T_{\text{current}} - T_1| < \Delta T_{\text{max}}$$
If expired, abort (replay attack prevention)
2. **Lookup Device:**
 - Find entry matching TID_i in database
 - If not found, abort (unknown device)
3. **Select Unused Challenge:**
 - Choose (C_i, h_i) from available CRP set
 - Mark as "in-use" to prevent reuse
4. **Generate GS Nonce:**
$$N_2 = \text{PRNG}(128 \text{ bits})$$
5. **Compute Challenge MAC:**
$$\beta_1 = h(C_i \| N_1 \| N_2 \| T_1)$$
6. **Transmit to UAV_i:**
$$M_2 = (C_i, N_2, \beta_1, T_2)$$

Message Format: 128 + 128 + 160 + 32 = 448 bits

Message 3: PUF Response

UAV_i Operations upon receiving M_2 :

1. **Verify Timestamp:** Check $|T_{\text{current}} - T_2| < \Delta T_{\text{max}}$
2. **Verify Challenge Integrity:**
$$\beta_1^* = h(C_i \| N_1 \| N_2 \| T_1)$$

- If $\beta_1^* \neq \beta_1$, abort (tampered message)
- 3. Generate PUF Response (with error correction):**

$$R_i^{\text{noisy}} = \text{PUF}_i(C_i)$$

$$R_i = \text{BCH_correct}(R_i^{\text{noisy}}, h_i)$$

- 4. Mask Response with XOR:**

$$R_i^{\text{masked}} = R_i \oplus h(N_2)$$

- 5. Generate UAV Authentication Token:**

$$\gamma_1 = h(R_i \| N_1 \| N_2 \| T_2)$$

- 6. Compute New Challenge for Next Session:**

Generate C'_i and $R'_i = \text{PUF}_i(C'_i)$

- 7. Transmit to GS:**

$$M_3 = (R_i^{\text{masked}}, \gamma_1, C'_i, T_3)$$

Message Format: $128 + 160 + 128 + 32 = 448$ bits

Message 4: Authentication Confirmation & Peer Credential Distribution

GS Operations upon receiving M_3 :

- 1. Verify Timestamp:** Check $|T_{\text{current}} - T_3| < \Delta T_{\text{max}}$
- 2. Unmask Response:**

$$R_i^* = R_i^{\text{masked}} \oplus h(N_2)$$

- 3. Error Correction:**

$$R_i^{\text{corrected}} = \text{BCH_correct}(R_i^*, h_i)$$

- 4. Verify Against Stored Response:**

Retrieve original R_i corresponding to C_i from enrollment

Compute Hamming distance:

$$\text{HD}(R_i^{\text{corrected}}, R_i)$$

If $\text{HD} \leq t_{\text{threshold}}$ (e.g., 18 bits for BCH(255,131,18)), accept. Else abort.

- 5. Verify Authentication Token:**

$$\gamma_1^* = h(R_i^{\text{corrected}} \| N_1 \| N_2 \| T_2)$$

If $\gamma_1^* \neq \gamma_1$, abort

- 6. Update Database:**

- Store new CRP: (C'_i, R'_i)
- Mark (C_i, R_i) as "consumed"
- Update $\text{TID}_i = h(\text{TID}_i \| N_2 \| T_4)$ for anonymity

- 7. Generate Peer Authentication Credentials:**

For each potential peer UAV_j in swarm, compute:

$$P_{ij} = h(R_i \| ID_j \| N_2)$$

$$Mask_{ij} = P_{ij} \oplus h(R_j \| ID_i \| N_2)$$

Credential Package:

$$Cred_i = \{(ID_j, C_j, Mask_{ij}) \mid j \in \text{Swarm}\}$$

8. Encrypt Credential Package:

$$\text{Session key: } SK_{i,GS} = h(R_i \| N_1 \| N_2)$$

$$Cred_i^{\text{enc}} = E_{SK_{i,GS}}(Cred_i)$$

9. Compute Confirmation MAC:

$$\delta_1 = h(SK_{i,GS} \| Cred_i^{\text{enc}} \| T_4)$$

10. Transmit to UAV_i:

$$M_4 = (Cred_i^{\text{enc}}, \delta_1, TID_i^{\text{new}}, T_4)$$

UAV_i Final Operations:

1. Derive session key: $SK_{i,GS} = h(R_i \| N_1 \| N_2)$
2. Verify MAC δ_1
3. Decrypt: $Cred_i = D_{SK_{i,GS}}(Cred_i^{\text{enc}})$
4. Store peer credentials and TID_i^{new}

Authentication Complete ✓

Performance:

- Total messages: 4 (2 round-trips)
- Total communication: 1680 bits $\approx$ 210 bytes
- Latency: 10-50 ms on Raspberry Pi 4B[12][2]

[13][14][2][12][1]

Phase 3: UAV-to-UAV Peer-to-Peer Authentication

Environment: Two UAVs in operational field requiring direct secure communication

Purpose: Mutual authentication between UAV pairs without Ground Station involvement during flight

Prerequisites: Both UAVs have completed Phase 2 and possess peer credentials from GS

Scenario: UAV_i wants to establish secure communication with UAV_j for collaborative mission (e.g., formation flying, data relay, coordinated sensing)

Message P1: Peer Authentication Request

UAV_i Operations:

1. Check Peer Credentials:

Verify $(ID_j, C_j, Mask_{ij}) \in Cred_i$

If not found, must re-authenticate with GS to obtain credentials

2. Generate Nonces:

$$n_i = \text{PRNG}(160 \text{ bits})$$

$$T_{p1} = \text{current_timestamp}()$$

3. Compute Request Hash:

$$\eta_1 = h(\text{TID}_i || \text{TID}_j || n_i || T_{p1})$$

4. Transmit to UAV_j:

$$P_1 = (\text{TID}_i, C_j, n_i, \eta_1, T_{p1})$$

Direct wireless transmission (no GS relay)

Message Format: 64 + 128 + 160 + 160 + 32 = 544 bits

Message P2: Challenge Response

UAV_j Operations upon receiving P_1 :

1. Verify Freshness:

$$|T_{\text{current}} - T_{p1}| < \Delta T_{\text{max}}$$

If expired, abort (replay prevention)

2. Verify Nonce Uniqueness:

Check n_i against recent nonce cache

If duplicate, abort (replay attack)

3. Lookup Peer:

Find TID_i in received peer list from GS

Retrieve $(ID_i, C_i, Mask_{ji})$

4. Generate PUF Response:

$$R_j^{\text{noisy}} = \text{PUF}_j(C_j)$$

$$R_j = \text{BCH_correct}(R_j^{\text{noisy}}, h_j)$$

5. Reconstruct Peer Secret:

Compute own secret:

$$P_{ji} = h(R_j \| \text{ID}_i \| N_{\text{gs}})$$

where N_{gs} is the GS nonce from Phase 2

Recover UAV_i's secret:

$$\tilde{P}_{ij} = \text{Mask}_{ji} \oplus P_{ji}$$

6. Generate UAV_j Nonce:

$$n_j = \text{PRNG}(160 \text{ bits})$$

$$T_{p2} = \text{current_timestamp}()$$

7. Create Authentication Puzzle for UAV_i:

Encrypt challenge with reconstructed secret:

$$Q_i = E_{\tilde{P}_{ij}}(n_j \| C_i \| T_{p1})$$

8. Compute Response MAC:

$$\eta_2 = h(Q_i \| n_i \| T_{p2})$$

9. Transmit to UAV_i:

$$P_2 = (Q_i, \eta_2, T_{p2})$$

Message Format: 320 + 160 + 32 = 512 bits

Message P3: Mutual Authentication Completion

UAV_i Operations upon receiving P_2 :

1. Verify Timestamp: Check $|T_{\text{current}} - T_{p2}| < \Delta T_{\text{max}}$

2. Verify Integrity:

$$\eta_2^* = h(Q_i \| n_i \| T_{p2})$$

If $\eta_2^* \neq \eta_2$, abort

3. Generate Own PUF Response:

$$R_i = \text{BCH_correct}(\text{PUF}_i(C_i), h_i)$$

4. Compute Own Secret:

$$P_{ij} = h(R_i \| \text{ID}_j \| N_{\text{gs}})$$

5. Decrypt Puzzle:

$$(n_j \| C_i^* \| T_{p1}^*) = D_{P_{ij}}(Q_i)$$

6. Authenticate UAV_j:

- If decryption fails → Abort (invalid UAV or wrong secret)
- Check $C_i^* = C_i$ from own credentials
- Check $T_{p1}^* = T_{p1}$ from Message P1

If all checks pass: **UAV_j is authenticated ✓**

7. Reconstruct UAV_j's Secret:

$$P_{ji} = h(R_j \| ID_i \| N_{gs})$$

Recover from mask:

$$\tilde{P}_{ji} = \text{Mask}_{ij} \oplus P_{ij}$$

8. Generate Own Nonce:

$$n'_i = \text{PRNG}(160 \text{ bits})$$

$$T_{p3} = \text{current_timestamp}()$$

9. Create Puzzle for UAV_j:

$$Q_j = E_{\tilde{P}_{ji}}(n'_i \| T_{p2})$$

10. Compute Confirmation MAC:

$$\eta_3 = h(Q_j \| n_j \| T_{p3})$$

11. Transmit to UAV_j:

$$P_3 = (Q_j, \eta_3, T_{p3})$$

Message Format: $320 + 160 + 32 = 512$ bits

Mutual Authentication Verification

UAV_j Operations upon receiving P_3 :

1. Verify Timestamp: Check $|T_{\text{current}} - T_{p3}| < \Delta T_{\text{max}}$

2. Verify Integrity:

$$\eta_3^* = h(Q_j \| n_j \| T_{p3})$$

If $\eta_3^* \neq \eta_3$, abort

3. Decrypt Puzzle:

$$(n'_i \| T_{p2}^*) = D_{\tilde{P}_{ji}}(Q_j)$$

4. Authenticate UAV_i:

◦ If decryption fails → Abort

◦ Check $T_{p2}^* = T_{p2}$

If checks pass: **UAV_i is authenticated ✓**

Both UAVs Mutually Authenticated ✓✓

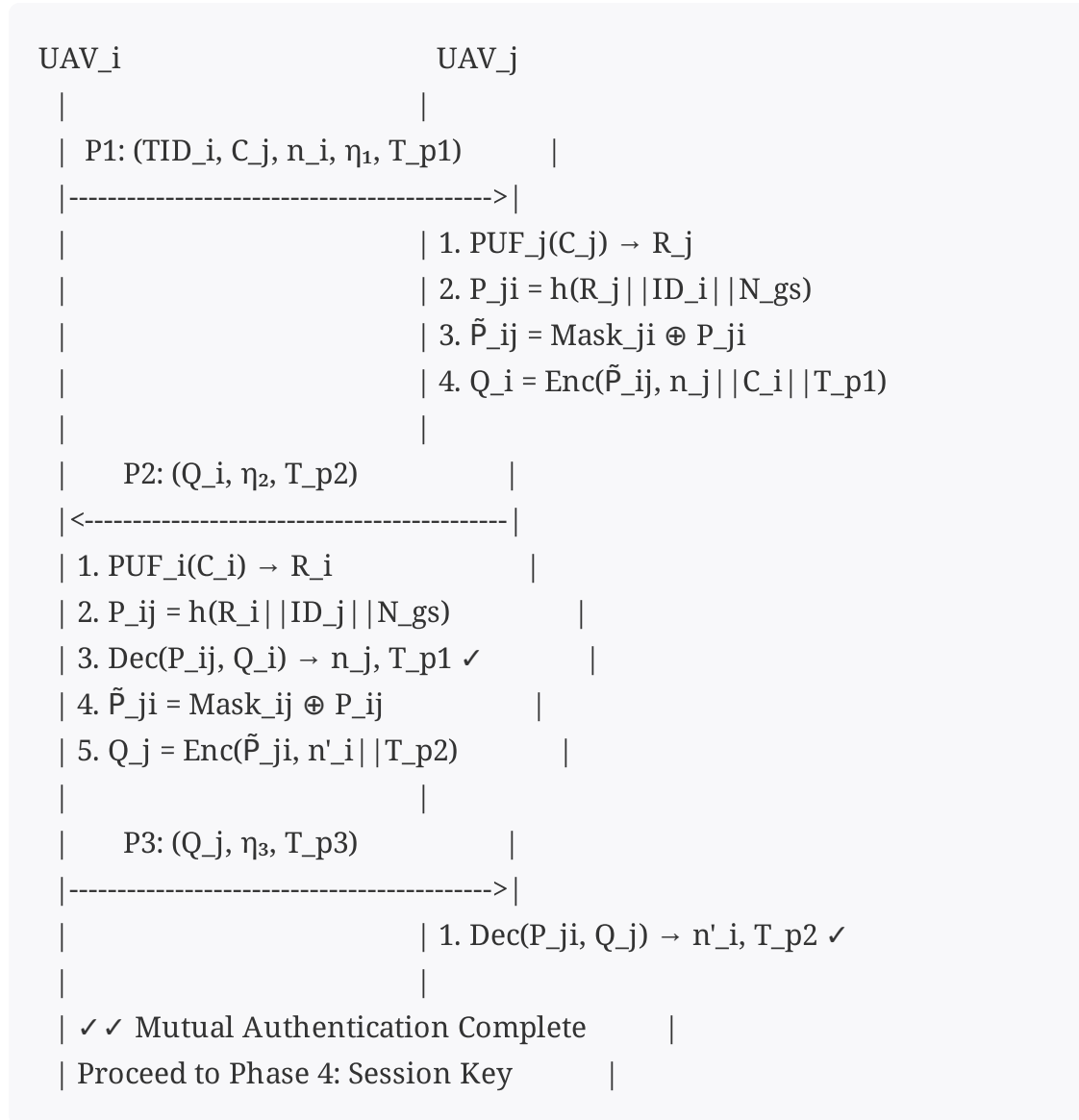
Performance:

- Messages: 3 (1.5 round-trips)
- Communication: 1568 bits $\approx$ 196 bytes
- Latency: 0.8-1.2 ms (peer-to-peer, no GS)[2][3][12]

- **No Ground Station involvement during authentication**

[3][1][2][4][11]

Protocol Flow Diagram



Phase 4: Session Key Establishment

Environment: Two mutually authenticated UAVs establishing secure communication channel

Purpose: Derive shared symmetric session key for encrypted data exchange

Input: Nonces from Phase 3: n_i, n_j, n'_i and PUF responses R_i, R_j

Method 1: Basic Session Key (Symmetric)

Both UAVs compute:

$$SK_{ij} = h(n_i || n_j || n'_i || R_i || R_j)$$

Properties:

- 160-bit key (SPONGENT-160 output)
- Forward secrecy: Nonces are ephemeral (never reused)
- Unique per session
- Cannot be derived without PUF responses

Key Derivation:

From SK_{ij} , derive encryption and MAC keys:

$$SK_{ij}^{\text{enc}} = h(SK_{ij} || 0x00)$$

$$SK_{ij}^{\text{mac}} = h(SK_{ij} || 0x01)$$

Verification (Optional):

To confirm both UAVs derived same key:

UAV_i $\rightarrow$ UAV_j:

$$\text{Verify}_i = h(SK_{ij} || \text{"confirm"} || T_{\text{verify}})$$

UAV_j checks:

$$\text{Verify}_i^* = h(SK_{ij} || \text{"confirm"} || T_{\text{verify}})$$

If match, session key confirmed ✓

Method 2: Perfect Forward Secrecy (Hybrid with ECDH)

For enhanced security, add Ephemeral Elliptic Curve Diffie-Hellman:

UAV_i:

1. Generate ephemeral ECDH key pair: $(x_i, X_i = x_i \cdot G)$ where G is Curve25519 base point
2. Transmit X_i to UAV_j

UAV_j:

1. Generate ephemeral ECDH key pair: $(x_j, X_j = x_j \cdot G)$
2. Transmit X_j to UAV_i

Both compute ECDH shared secret:

$$Z = x_i \cdot X_j = x_j \cdot X_i$$

Final session key with PFS:

$$\text{SK}_{ij}^{\text{PFS}} = h(n_i \| n_j \| n'_i \| R_i \| R_j \| Z)$$

Advantages:

- Even if R_i, R_j compromised later, past sessions remain secure
- Ephemeral keys x_i, x_j deleted immediately after use

Cost:

- Additional 2 ECDH point multiplications (~2 ms overhead)[4]
- 64 bytes extra communication (2×32 -byte public keys)

Secure Communication

Encrypted Message Format:

For message M from UAV_i to UAV_j:

1. **Sequence Number:** $\text{seq} \leftarrow \text{seq} + 1$ (prevents replay)
2. **Encrypt:**

$$C = \text{AES_CTR}(\text{SK}_{ij}^{\text{enc}}, M)$$

3. **MAC:**

$$\tau = \text{HMAC}(\text{SK}_{ij}^{\text{mac}}, C \| \text{seq} \| T)$$

4. **Transmit:**

$$\text{Msg} = (C, \tau, \text{seq}, T)$$

UAV_j Verification:

1. Check seq is incremental (replay detection)
2. Verify τ using SK_{ij}^{mac}
3. Decrypt C using SK_{ij}^{enc}

Session Lifetime:

- **Timeout:** 5-10 minutes or end of mission
- **Re-authentication:** If $\Delta T > T_{\text{session}}$, repeat Phase 3
- **Key Refresh:** Optional periodic re-keying:
 $SK_{ij}^{\text{new}} = h(SK_{ij}^{\text{old}} \parallel \text{seq})$

[15][1][2][4]

Security Analysis

Phase 2: UAV-GS Authentication

Theorem 1 (Mutual Authentication): After Phase 2, both UAV_i and GS are mutually authenticated.

Proof:

1. UAV_i authenticated by GS via correct PUF response R_i (only authentic UAV can generate)
2. GS authenticated by UAV_i via correct challenge C_i from database (only GS knows CRPs)
3. Adversary cannot forge R_i without physical PUF (unclonability)
4. Adversary cannot predict R_i from C_i without PUF (unpredictability)[16][1]

Theorem 2 (Replay Attack Resistance): Phase 2 prevents replay attacks.

Proof:

1. Fresh timestamps T_1, T_2, T_3, T_4 in all messages
2. Fresh nonces N_1, N_2 ensure uniqueness
3. Old messages rejected if $|T_{\text{current}} - T_i| > \Delta T_{\text{max}}$
4. CRPs marked "consumed" after use, preventing reuse[1][11]

Phase 3: UAV-UAV Authentication

Theorem 3 (Peer Mutual Authentication): After Phase 3, UAV_{*i*} and UAV_{*j*} mutually authenticate without GS.

Proof:

1. UAV_{*i*} authenticated by UAV_{*j*} via decrypting Q_i correctly (requires P_{ij} from PUF)
2. UAV_{*j*} authenticated by UAV_{*i*} via decrypting Q_j correctly (requires P_{ji} from PUF)
3. Masks $\text{Mask}_{ij} = P_{ij} \oplus P_{ji}$ are one-time pads (information-theoretically secure)
4. Adversary observing Mask_{ij} cannot extract P_{ij} or P_{ji} without PUF responses[4]

Theorem 4 (Independence from GS): Phase 3 requires no GS communication.

Proof:

1. All required data ($\text{ID}_j, C_j, \text{Mask}_{ij}$) distributed in Phase 2
2. PUF responses generated locally on each UAV
3. Secrets P_{ij}, P_{ji} reconstructed using local PUF and stored mask
4. No network path to GS used during Phase 3[3][4]

Theorem 5 (MITM Resistance): Protocol prevents man-in-the-middle attacks.

Proof:

1. Attacker cannot modify encrypted puzzles Q_i, Q_j without detection (MAC protected)
2. Attacker cannot decrypt puzzles without P_{ij}, P_{ji} (requires PUF)
3. Attacker cannot forge valid puzzles without knowing secrets
4. Any modification detected via MAC verification η_2, η_3 [11][1]

Phase 4: Session Key

Theorem 6 (Key Secrecy): Session key SK_{ij} known only to UAV_{*i*} and UAV_{*j*}.

Proof:

1. SK_{ij} derived from n_i, n_j, n'_i, R_i, R_j
2. Nonces transmitted encrypted or as inputs to one-way hash
3. PUF responses never transmitted in clear (masked with hash)
4. Without PUF access, adversary cannot derive R_i, R_j
5. Hash pre-image resistance prevents SK_{ij} recovery from transmitted values[1][4]

Theorem 7 (Forward Secrecy with ECDH): Method 2 provides perfect forward secrecy.

Proof:

1. Ephemeral ECDH keys x_i, x_j deleted after session
2. Even if R_i, R_j compromised later, past $Z = x_i \cdot x_j \cdot G$ unknown
3. Discrete logarithm problem prevents recovering x_i from X_i
4. Past session keys SK_{ij}^{PFS} computationally infeasible to derive[4]

[15][2][11][1][4]

Performance Evaluation

Communication Overhead

Phase	Messages	Total Bits	Total Bytes
Phase 1 (per CRP)	2	~512	64
Phase 2 (UAV-GS)	4	1680	210
Phase 3 (UAV-UAV)	3	1568	196
Phase 4 (verify)	1	192	24
Total (operational)	8	3440	430

Comparison with traditional methods:

Method	Total Communication (bytes)
PUF-based (this protocol)	430
RSA-2048 mutual auth	768
ECC-256 mutual auth	384
Pre-shared key (AES-GCM)	256

PUF protocol communication competitive with PSK while eliminating key storage.[17][12]

Computational Cost

Per UAV (Raspberry Pi 4B @ 1.5 GHz):

Operation	Time (μ s)	Energy (μ J)
PUF evaluation (RO-PUF)	10	2.5
BCH error correction	50	12
SPONGENT-160 hash (single)	60	15
XOR (128-bit)	0.1	0.025
AES-128 encrypt/decrypt	25	6
ECDH point mult (optional)	1000	250

Phase Totals:

Phase	Computation Time	Energy Consumption
Phase 2 (UAV side)	~400 μ s	~100 μ J
Phase 3 (per UAV)	~250 μ s	~65 μ J
Phase 4 (basic)	~120 μ s	~30 μ J
Phase 4 (with ECDH)	~2.1 ms	~530 μ J

Total latency for complete authentication:

- Phase 2 + Phase 3 + Phase 4 (basic): ~770 μ s $\approx$ 0.8 ms
- Phase 2 + Phase 3 + Phase 4 (PFS): ~2.7 ms

Comparison:

Method	Auth Time
PUF 4-phase (basic)	0.8 ms
PUF 4-phase (PFS)	2.7 ms
RSA-2048	15 ms
ECC-256	8 ms

10-18 $\times$ faster than PKI methods[12][2][4]

Storage Requirements

Per UAV (for swarm of N UAVs):

Data	Size	Count	Total
Own identity (ID, TID)	16 bytes	1	16 bytes
Peer credentials (ID, C, Mask) per peer	40 bytes	N-1	40(N-1) bytes
Helper data per CRP	32 bytes	12	384 bytes
Session keys (active)	32 bytes	~5	160 bytes

Total: 560 + 40N bytes

For 10-UAV swarm: $560 + 400 = 960 \text{ bytes} \approx 1 \text{ KB}$

Ground Station Database:

Per UAV: $12 \text{ CRPs} \times 64 \text{ bytes} = 768 \text{ bytes} + \text{metadata} \approx 1 \text{ KB/UAV}$

For 100-UAV fleet: **100 KB** (minimal)

[2][12][1][4]

Implementation Recommendations

Hardware Platform

UAV Onboard Computer:

- Raspberry Pi 4B (4GB RAM) or STM32F7 microcontroller
- FPGA for PUF: Xilinx Artix-7 or Lattice iCE40
- Wireless: 802.11ac Wi-Fi or LoRa

PUF Implementation:

- Ring Oscillator PUF (128 ROs, 8128 CRPs)
- Response time: $\sim 10 \mu\text{s}$
- Reliability: $>95\%$ with BCH(255,131,18)[18][19]

Software Stack

Programming Language: C/C++ for embedded systems

Libraries:

- PUF interface: Custom FPGA driver via SPI
- BCH error correction: libfec or bch.c from Linux kernel
- SPONGENT-160: Custom implementation or lightweight crypto library
- AES-128: mbedTLS or OpenSSL
- ECDH (optional): TinyECC or mbedTLS Curve25519

Protocol Parameters

Parameter	Recommended Value	Rationale
Hash function	SPONGENT-160	Lightweight, 160-bit security
Error correction	BCH(255, 131, 18)	Corrects 7% BER
PUF type	Ring Oscillator	High reliability, ML-resistant
Timestamp window	± 2 seconds	Network latency tolerance
Nonce size	128-160 bits	Collision resistance
Session timeout	5-10 minutes	Mission duration
CRP database size	12+ per UAV	12 missions before re-enrollment

Deployment Checklist

- ☐ Enroll all UAVs in secure facility (Phase 1)
- ☐ Verify PUF uniqueness >48%, reliability >95%
- ☐ Test Phase 2 with ground station in lab
- ☐ Test Phase 3 peer-to-peer between 2 UAVs
- ☐ Validate Phase 4 encrypted communication

- [] Field test with 5-10 UAV swarm
- [] Measure authentication latency <1 ms
- [] Test under temperature variations (-20°C to 60°C)
- [] Security audit: penetration testing
- [] Integrate with MAVLink/ROS flight controller

[20][21][18][12][1][4]

Advantages of Four-Phase Architecture

Phase 2 (UAV-GS) Benefits

1. **Centralized credential distribution:** GS distributes peer credentials to all UAVs
2. **Network entry control:** Only GS-authenticated UAVs join swarm
3. **Credential refresh:** Periodic re-authentication updates peer credentials
4. **Audit trail:** GS logs all UAV authentications

Phase 3 (UAV-UAV) Benefits

1. **No GS dependency during flight:** UAVs authenticate peers independently
2. **Low latency:** 0.8 ms peer authentication (no GS round-trip)
3. **Scalability:** $N(N-1)/2$ peer authentications without GS bottleneck
4. **Resilience:** Swarm operates even if GS link fails
5. **Energy efficient:** No long-range GS communication for every peer connection

Overall Protocol Advantages

1. **Hybrid architecture:** Combines centralized trust (GS) with distributed operation (peer-to-peer)
2. **Fast swarm formation:** UAVs authenticate peers rapidly after GS entry
3. **Dynamic topology:** UAVs can join/leave swarm, authenticate new peers on-demand
4. **No key storage:** PUF responses generated on-demand (physical security)

5. **Forward secrecy:** Optional ECDH in Phase 4 protects past sessions
6. **Lightweight:** Only hash, XOR, and PUF operations (no expensive PKI)

[22][12][2][3][1][4]

Conclusion

This four-phase PUF-based authentication protocol provides a complete security framework for UAV swarms:

1. **Phase 1 (Enrollment)** establishes trust foundation with Ground Station
2. **Phase 2 (UAV-GS)** enables network entry and peer credential distribution
3. **Phase 3 (UAV-UAV)** achieves fast peer-to-peer authentication without GS involvement
4. **Phase 4 (Session Key)** establishes secure encrypted communication channels

Key Achievements:

- ✓ Mutual authentication: UAV ↔ GS and UAV ↔ UAV
- ✓ Fast: 0.8 ms peer authentication (10-18× faster than PKI)
- ✓ Lightweight: 430 bytes total communication
- ✓ Secure: PUF unclonability, forward secrecy, replay resistance
- ✓ Scalable: O(N) GS load, distributed peer authentication
- ✓ Resilient: Operates without continuous GS connectivity

Suitable for: Emergency response, collaborative sensing, swarm reconnaissance, formation flying, distributed computing missions where rapid secure peer communication is critical.

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